

Saadat Rafid Ahmed

Lecturer, Department of Computer Science & Engineering, BRAC University, Dhaka

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Academic Profile

Lecturer and researcher in Computer Science with a focus on **low-resource NLP**, **adversarial machine learning**, and **AI for education**. Currently teaching 300+ students per semester across ML, NLP, and Numerical Methods while supervising six undergraduate thesis groups. Concurrently leading an ML team building RAG-based educational AI systems serving 155k+ students. Research explores the robustness of transfer learning in Bengali, with published work on adversarial attacks and sentiment analysis. Seeking fully-funded graduate opportunities to deepen research in trustworthy AI and language technology for underrepresented languages.

Research Interests

Natural Language Processing for Low-Resource Languages · Adversarial Machine Learning · Retrieval-Augmented Generation · Bengali Language Technology · Transfer Learning Robustness · AI for Education · ML Fairness & Bias

Education

BSc in Computer Science & Engineering · BRAC University 2020 – 2023
CGPA: **4.00 / 4.00** · 100% merit-based scholarship · Vice-Chancellor's Award (7 semesters)
Thesis: *destroR: Attacking Transfer Models with Obfuscated Examples to Discard Perplexity*
Advisor: Dr. Farig Yousuf Sadeque, Assistant Professor, BRAC University
Selected Coursework: Machine Learning, Neural Networks & Deep Learning, Natural Language Processing, Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Signal Processing

Publications & Preprints

* denotes first/corresponding author

- [1] **S. R. Ahmed***, “destroR: Attacking Transfer Models with Obfuscated Examples to Discard Perplexity,” *arXiv preprint arXiv:2511.11309*, 2025. [PDF]
- [2] **S. R. Ahmed***, “BSenti: A Comprehensive Bangla Sentiment Classification Library,” Open-source Python library and benchmark. Unifies fragmented Bangla sentiment datasets; provides comparative evaluation across statistical (SVM, NB, LR) and deep-learning (BiLSTM, BERT) architectures.
- [3] **S. R. Ahmed***, “SemEval 2023 Task 10: Explainable Detection of Online Sexism (EDOS),” *Proceedings of SemEval-2023, co-located with ACL 2023*. Ranked **Top 12 worldwide** out of 80+ teams; Micro F1 of 53.47%.

Research Experience

Thesis Research — Adversarial NLP for Bengali · BRAC University 2022 – 2023
Advisor: Dr. Farig Yousuf Sadeque

- Pioneered integration of Bengali into adversarial attacks and adversarial training, opening a new research direction for low-resource NLP security
- Devised **three novel attack strategies** exploiting Bengali morphology at character and word levels: homograph substitution, suffix perturbation, and semantic-preserving word replacement
- Achieved **18–40% F1 score degradation** across BERT-based and LSTM-based classification models, demonstrating fragility of transfer-learned Bengali representations
- First author; manuscript under review at an international venue

Thesis Supervision — 6 Undergraduate Groups · BRAC University May 2024 – Present

- **Track A — AI-assisted Legal Aid:** NLP-driven document analysis for legal accessibility; information extraction from Bengali legal texts and automated case summarization
- **Track B — User Behavior & Update Systems:** ML-based engagement modeling, feedback pipelines, and behavioral analytics for educational platforms
- Research themes span NLP, ML Bias & Ethics, Reinforcement Learning, and AI in Health & Education

Applied Research — RAG & Educational AI · Innospace Infotech Ltd. Nov 2024 – Present

- Designed Hierarchical RAG architectures with BM25 reranking for Bangladesh's national curriculum (HSC/SSC), enabling context-aware multi-turn dialogue for 155k+ students
- Developed HRAG validation pipeline using multi-hop reasoning for automated QA on 170k+ questions; built OCR + T5 pipeline for mixed Bengali-English document extraction

Teaching Experience

- Lecturer, Department of CSE · BRAC UniversityMay 2024 – Present
- Sole course designer for **CSE440**: Natural Language Processing II — Labs; developed full syllabus, assignments, and lab content from scratch
 - Teaching **300+ students/semester** across:
 - CSE422** — Introduction to Machine Learning / Artificial Intelligence
 - CSE330** — Numerical Methods | **CSE440** — NLP II Labs | **MAT120** — Discrete Mathematics
 - Designed original lab exercises: **Polynomial Interpolation** (Lagrange, Hermite, Newton, Chebyshev nodes) and **Neural Network** (backpropagation, gradient descent, optimization)
 - Created comprehensive LaTeX Beamer slide decks and structured lecture notes with university-branded formatting

- Undergraduate Teaching Assistant · BRAC UniversityOct 2022 – Dec 2023
- Conducted lab sessions and led study groups for 300+ students in CSE220 (Data Structures), MAT215 (Mathematics for ML & Signal Processing), PHY112 (Principles of Physics II)

Service to the Department

- Volunteer Web Systems Development · BRAC University, Dept. of CSEMay 2024 – Present
- Designed, developed, and maintained three internal platforms for the department:
- BRACU CSE SDS Portal** — Faculty management system centralizing admin forms, resource requests, and workflows
 - Thesis Group Management System** — End-to-end thesis lifecycle — registration, advisor assignment, proposal tracking, panel scheduling, and final mark submission
 - Peer Evaluation System** — Anonymous faculty peer-review with LLM-based summarizer preserving reviewer anonymity

Industry Experience

- ML Team Lead · Innospace Infotech Ltd.Nov 2024 – Present
- Leading AI/ML team building educational technology for [Lecture Publications Ltd.](#) and the [Britto App](#) (155k+ students). Delivered QA Bot, Question Splitter, RAG Chatbot, and Analytics Platform. Designed ETL pipelines reducing costs by 52%; Transformer POS tagger (3× speed); student tracking (+23% engagement).
- Data Scientist · Robi Axiata Ltd.Mar 2024 – May 2024
- Built churn prediction model on 47M users’ transactional data (170 TB) and a user-facing AI chatbot with chain-of-thought reasoning alongside an HR-facing RAG system for internal query resolution.

Technical Proficiency

Languages	Python, C, C++, TypeScript, JavaScript, Dart, WolframScript
ML & NLP	Scikit-learn, HuggingFace Transformers, TensorFlow, TextAttack, NLTK, LangChain, vLLM
RAG & Vector Stores	ChromaDB, Qdrant, BM25, Hierarchical RAG, Multi-hop Reasoning
MLOps & Cloud	AWS (SageMaker, Redshift), Docker, Kubernetes, Kubeflow, Paperspace
Data Engineering	ETL Pipelines, PostgreSQL, MongoDB, BigQuery, Metabase
Frameworks	Flask, FastAPI, GraphQL, Flutter, CrewAI, GROQ, MistralAI

Awards & Honors

- 100% Merit-Based Scholarship, BRAC University2020 – 2023
- Vice-Chancellor’s Award (7 semesters) — perfect 4.00 CGPA each semester2020 – 2023
- Top 12 Worldwide — SemEval 2023 Task 10 (EDOS), co-located with ACL 20232023

Service & Leadership

- Lead Software Research Member, Robotics & Intelligent Systems (RIS), BRAC UniversityDec 2021 – Dec 2023
- Led software development for robotics projects; mentored junior members on embedded systems
- Senior Executive, IT Department, BRAC University Adventure Club2021 – 2023

References

Dr. Farig Yousuf Sadeque
Assistant Professor, Dept. of CSE, BRAC University
Thesis Advisor · farig.sadeque@bracu.ac.bd

Languages: Bengali (native), English (fluent) · Interests: Guitar, Harmonium, Musical composition